

# Sequence Diagram vs System Sequence Diagram

# Sequence Diagram (Detailed Interaction View)

## Definition

A Sequence Diagram represents how objects interact with each other over time within a system. It shows the sequence of messages exchanged between objects to accomplish a functionality.

## Key Characteristics

- Focuses on internal system behavior
- Shows multiple objects/classes
- Includes:
  - Lifelines (objects)
  - Messages (method calls)
  - Activation bars (execution)
- Time flows top to bottom

# Purpose of Sequence Diagram

## Purpose

- To design and understand internal logic
- To map methods, responsibilities, and collaborations
- Used during detailed design phase

## Example

In an online shopping system:

- Customer → Cart → PaymentService → Database

All interactions are shown step-by-step.

# System Sequence Diagram (High-Level Interaction View)

## Definition

A System Sequence Diagram (SSD) shows the interaction between an external actor and the system as a whole.

## Key Characteristics

- Treats the system as a black box
- Only two main elements:
  - Actor (user or external system)
  - System (single entity)
- Shows system-level events (inputs/outputs)

# Purpose of SSD

## Purpose

- To define system behavior from user perspective
- To identify system operations
- Used during requirement analysis phase

## Example

In the same shopping system:

- Customer → System: addItem()
- Customer → System: checkout()

(No internal components like Cart or PaymentService are shown)

# Core Differences

| Aspect          | Sequence Diagram             | System Sequence Diagram                    |
|-----------------|------------------------------|--------------------------------------------|
| Level of Detail | Detailed                     | High-level                                 |
| Focus           | Internal object interactions | Actor–system interaction                   |
| System View     | White-box                    | Black-box                                  |
| Elements        | Multiple objects/classes     | Single system + actor<br>+ External system |
| Usage Phase     | Design phase                 | Requirement analysis                       |
| Complexity      | High                         | Simple                                     |
| Purpose         | Design logic                 | Define system operations                   |

A System Sequence Diagram is conceptually a precursor to a full Sequence Diagram:

- SSD identifies what operations the system must support
- Sequence Diagram explains how those operations are internally implemented

## Key Principle

Separation of concerns between external behavior and internal design

# When to Use Which

## Use SSD when:

- Defining use cases
- Communicating with non-technical stakeholders
- Capturing functional requirements

## Use Sequence Diagram when:

- Designing class interactions
- Implementing logic in code
- Debugging or optimizing workflows

# Conceptual Summary

- SSD = What the system does
- Sequence Diagram = How the system does it